Deep Learning

(01076533)

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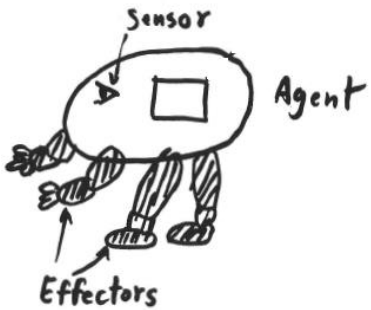
Some slides are adapted from
- cs231n: Convolutional Neural Networks for Visual Recognition @Stanford University

Types of learning



clustering

1. Unsupervised learning

- The agent learns patterns in the input even though no explicit label is supplied.
- The most common unsupervised learning task is clustering.
- For example, a taxi agent might gradually develop a concept of "Good traffic days" and "Bad traffic days" without ever being given labeled examples of each by a teacher.

  - X1: Rain, BigEvent, WorkDay, MonthEnd
  - X2: NonRain, NonBigEvent, NonWorkDay, NonMonthEnd
  - X3: Rain, NonBigEvent, WorkDay, NonMonthEnd
  - X4: NonRain, BigEvent, NonWorkDay, MonthEnd

A taxi agent may cluster X1, X4 as one cluster and X2, X3 as another cluster.

learning policy เป็นการอบรมให้ learn ไปเรื่อย

2. Reinforcement learning

- The agent learns from a series of reinforcements – rewards or punishments. เรียนรู้โดยให้คะแนนถ้าทำถูกตามกฎ ถ้าทำไม่ถูกไม่ได้คะแนน
- For example, the lack of a tip at the end of the journey gives the taxi agent an indication that it did something wrong.
- It is up to the agent to decide which of the actions prior to the reinforcement were most responsible for it.

3. Supervised learning

ประเภท classification

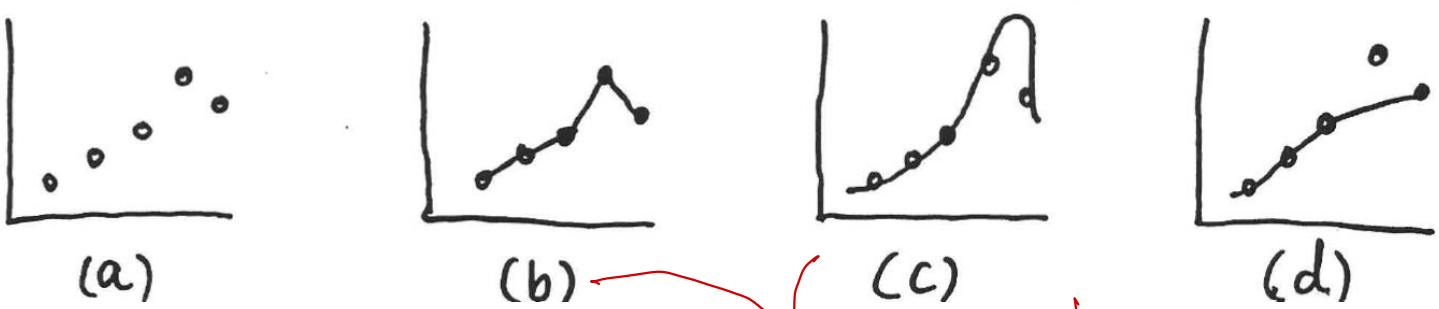
- The agent observes some example input-output pairs and learns a function that maps from input to output.

- For example, from a taxi agent problem. By using given labeled "input-output" examples, a taxi agent will learn a function that maps form input to the outputs "Good traffic days" and "Bad traffic days".

  - X1: Rain, BigEvent, WorkDay, MonthEnd, "Bad traffic days"
  - X2: NonRain, NonBigEvent, NonWorkDay, NonMonthEnd, "Good traffic days"
  - X3: Rain, NonBigEvent, WorkDay, NonMonthEnd, "Good traffic days"
  - X4: NonRain, BigEvent, NonWorkDay, MonthEnd, "Bad traffic days"

Overfitting learn nosie มากเกิน

- Overfitting occurs when a statistical learning model describes random errors or noises instead of the underlying relationship.
  - A learning model that has been **overfit** will generally have poor predictive performance, as it can exaggerate minor fluctuations in the data.
  - A good learning model should be **generalize** to both unseen and training data.



overfit

generalize ดีกว่า

generalize ดีกว่า

Fig. a : Training data

Fig. b and c tend to overfit the training data

Fig. d tends to generalize the training data (good hypothesis)

การวัดว่า medel ไหนดีกว่า ดู error rate

Evaluating the hypothesis

- We can simply use the error rate to check the quality of the hypothesis.

การ test ที่ model ไม่เคยเห็นมาก่อน
จะดีกว่า น่าเชื่อถือกว่า

- $\text{Error rate} = \dfrac{\text{Number of examples which are wrong predicted}}{\text{Total number of examples}}$

- However, a hypothesis h has a low error rate on the training set does not mean that it will generalize well.

- To get the more accurate evaluation, we need to test the hypothesis on an unseen set (test set) of examples.

การแบ่ง dataset สำหรับ Test หรือ train

- **Holdout cross-validation** :

เหมาะกับ Big Dataset

มีการกระจายตัวของข้อมูลเยอะ ทำให้ผลที่ได้ออกมาใกล้เคียงกัน

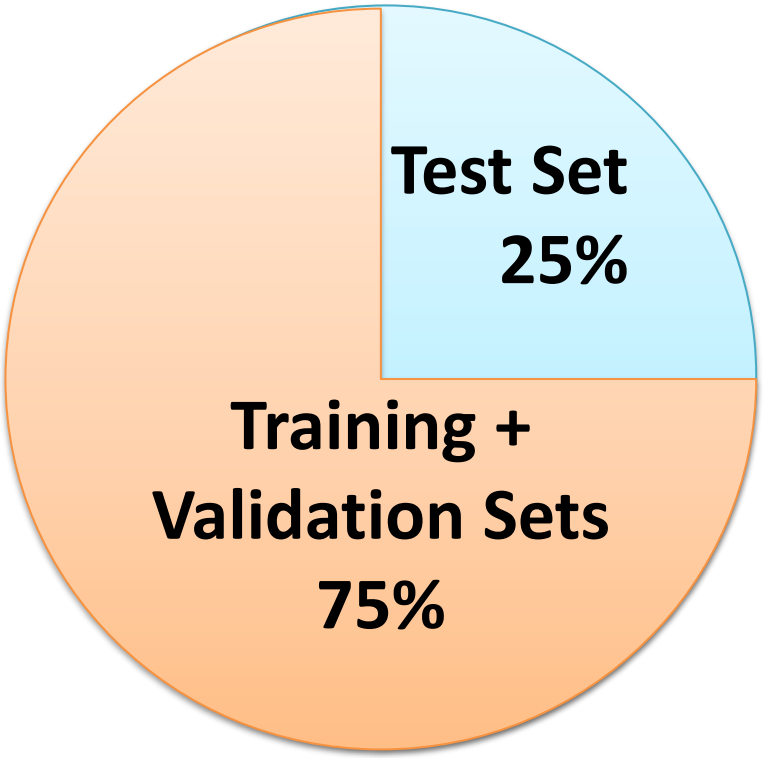
Randomly split the available data into a training set (e.g. 75%) and a test set (e.g. 25%).

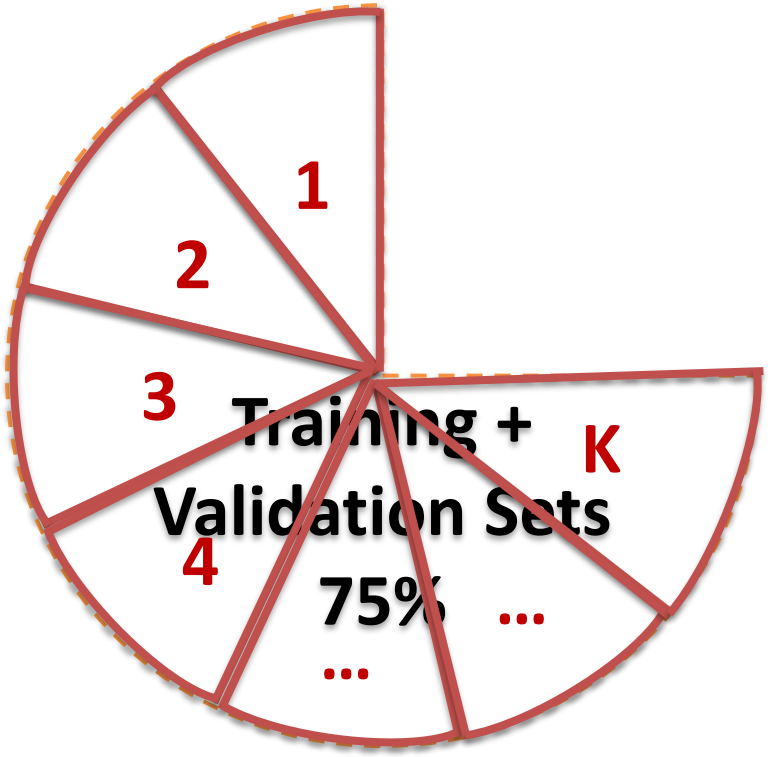
- **K-fold cross-validation**:

เหมาะกับ Low-meduim Dataset

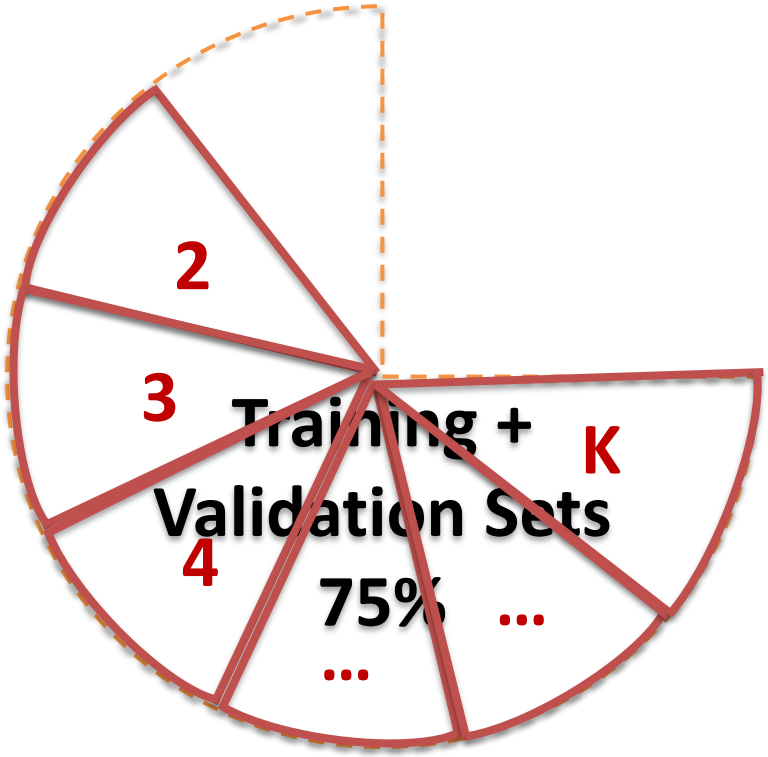
  - Split the training data into k equal subsets.
  - Then perform k rounds of learning. On each round, 1/k of the data is held out as a validation set and the remaining examples are used as training data.
  - The average error rate of the k rounds is finally calculated. Popular values for k are 5 and 10. The extreme is k=n (the total number of examples), also known as leave-one-out cross-validation or LOOCV.

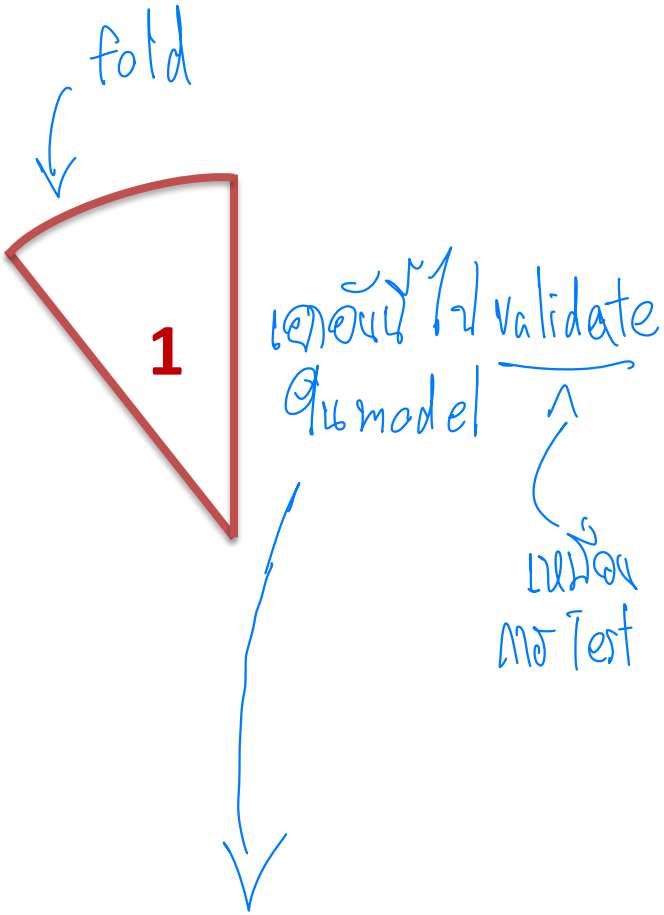
K-Fold Cross Validation





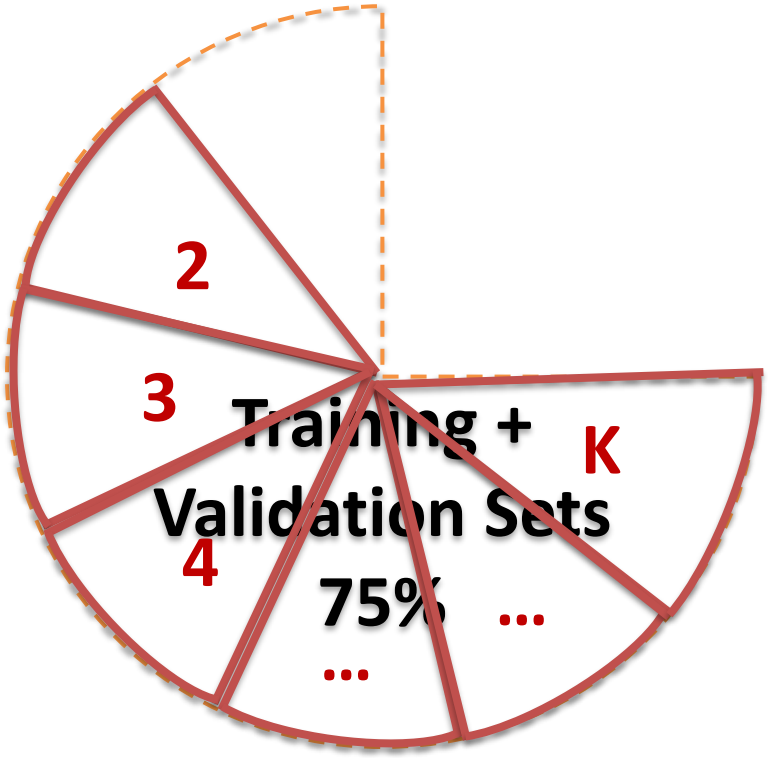
Learning Model

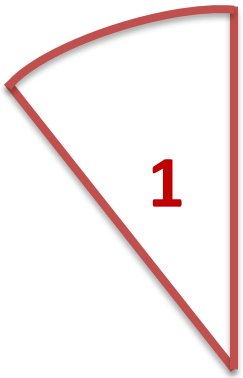




Learning Model

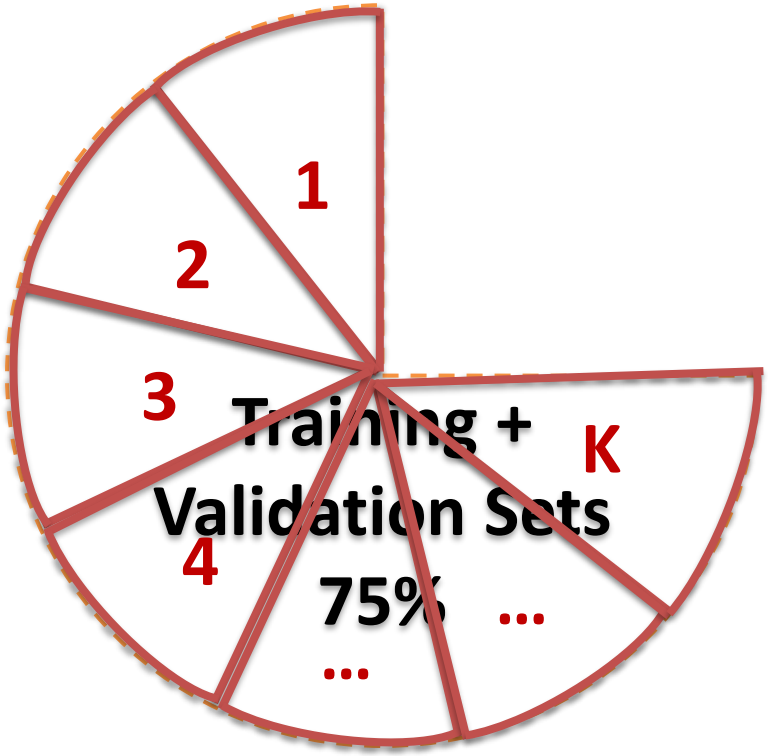
Training1 Accuracy/Error





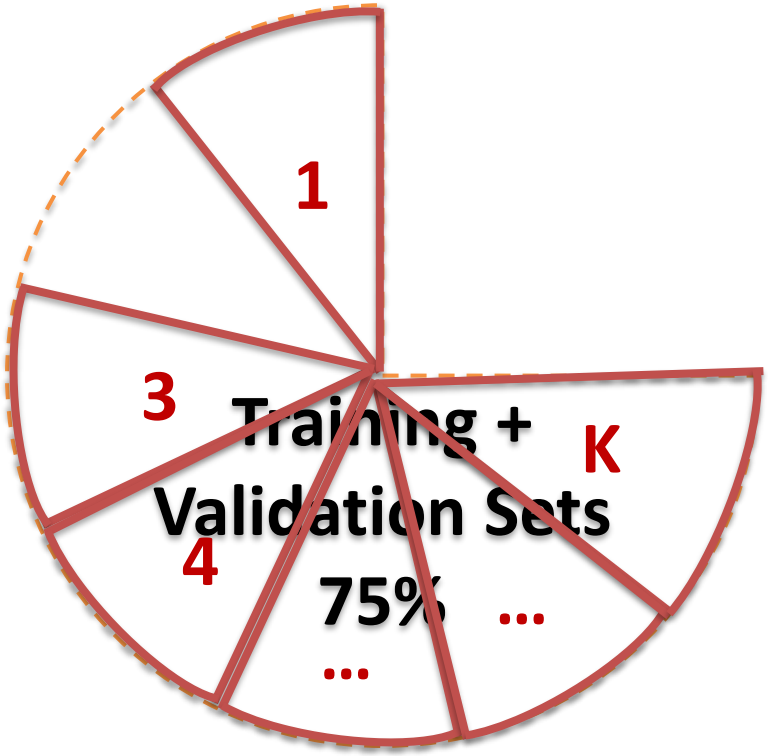
Learning Model

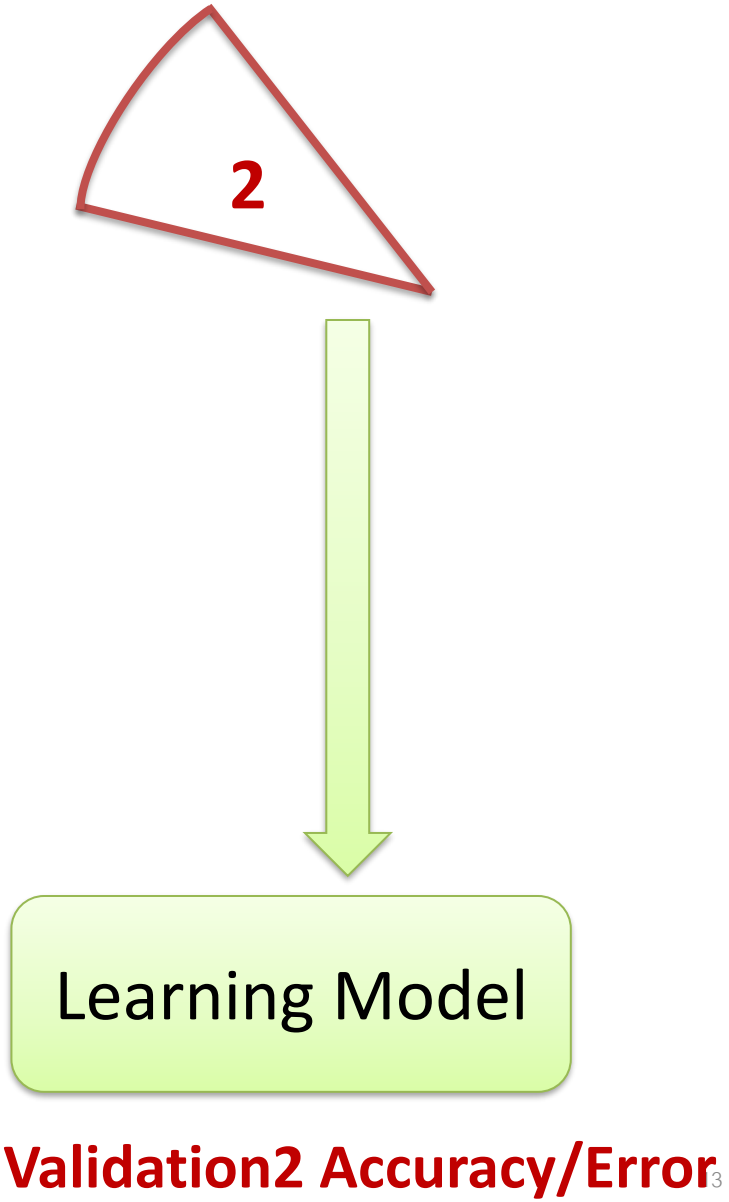
Validation1 Accuracy/Error

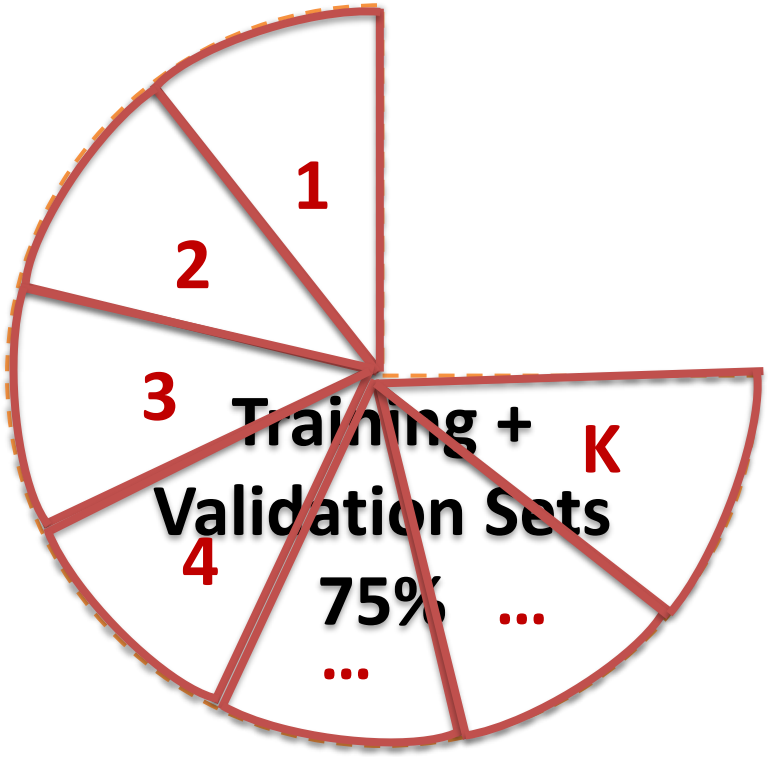


Learning Model

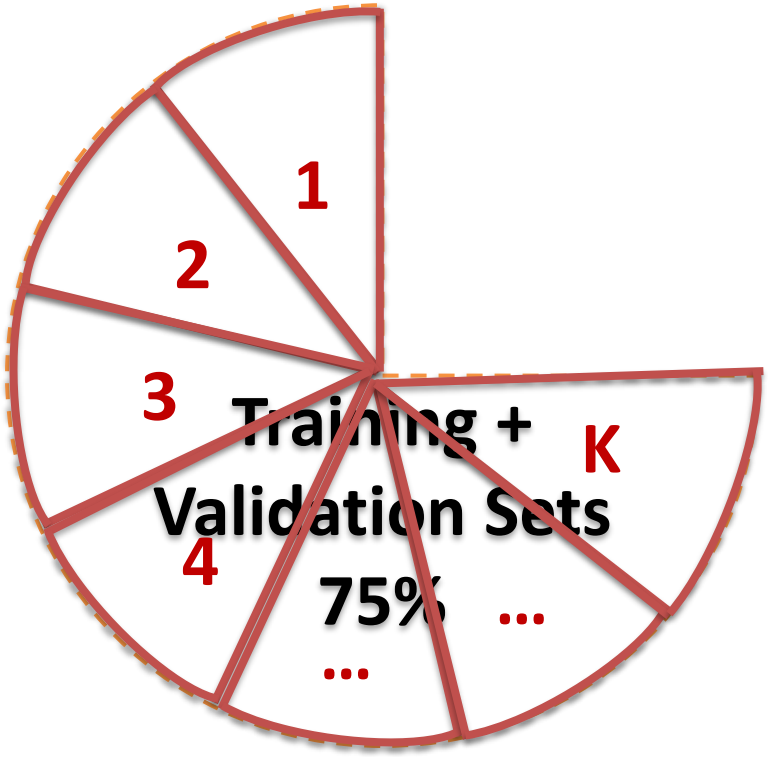
Training2 Accuracy/Error


1
3
Training +
Validation Sets
K
4
75%
...
...

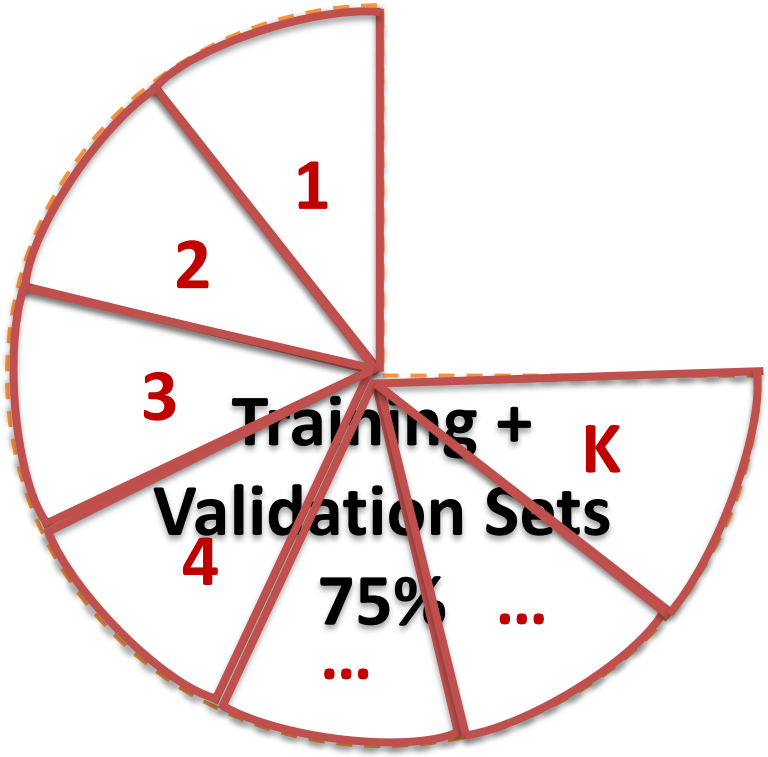

2
Learning Model
Validation2 Accuracy/Error



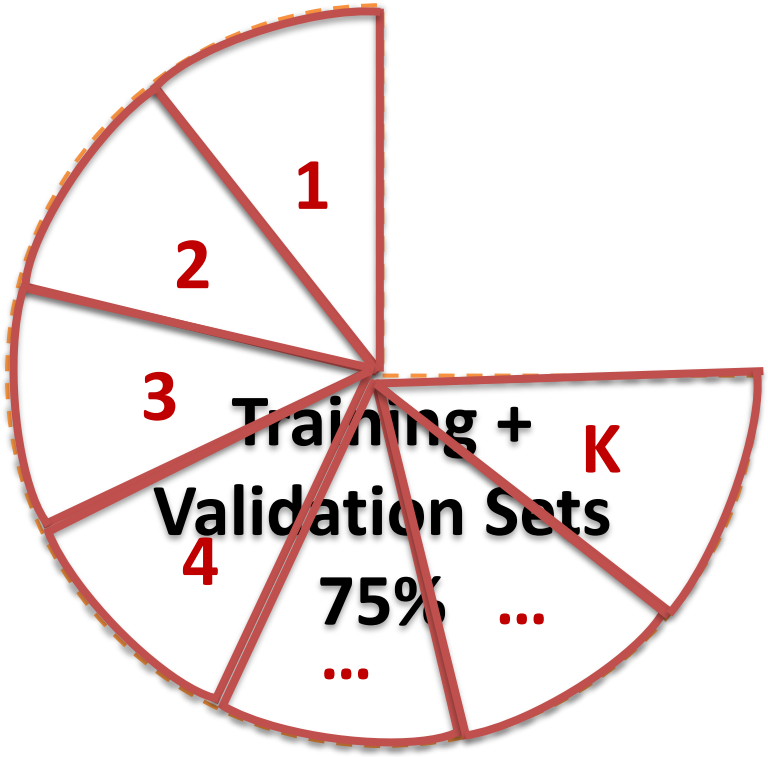
Learning Model



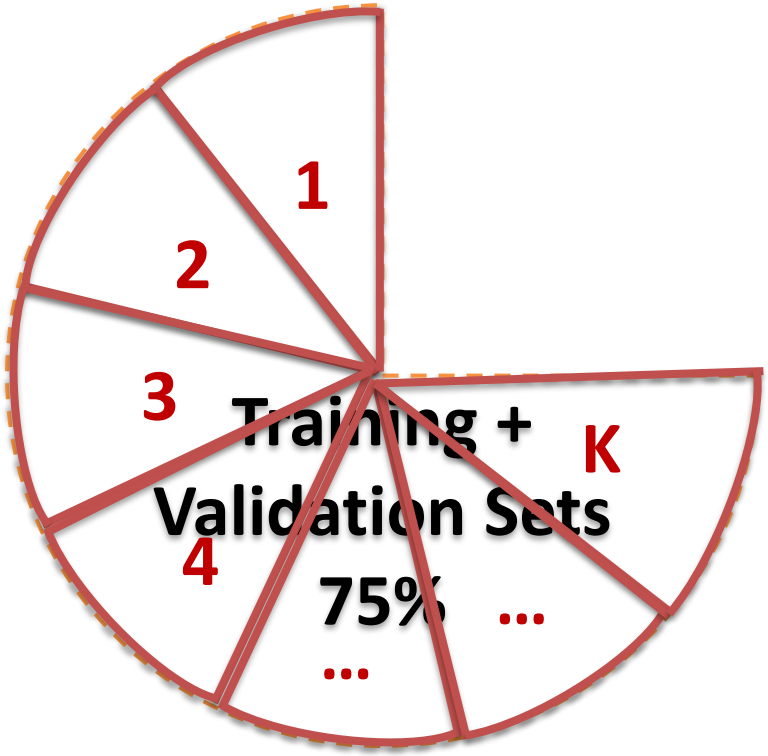
Learning Model



Learning Model

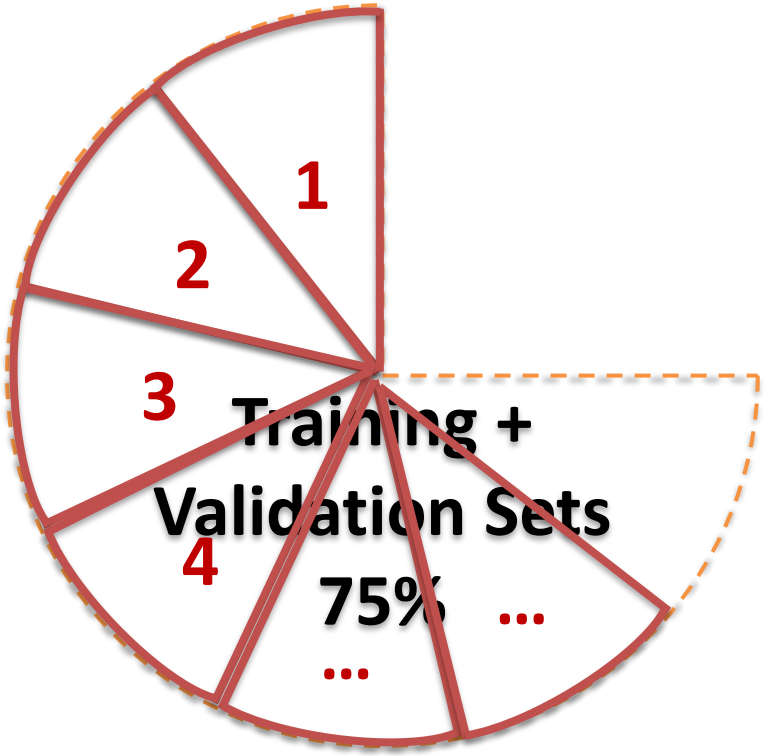


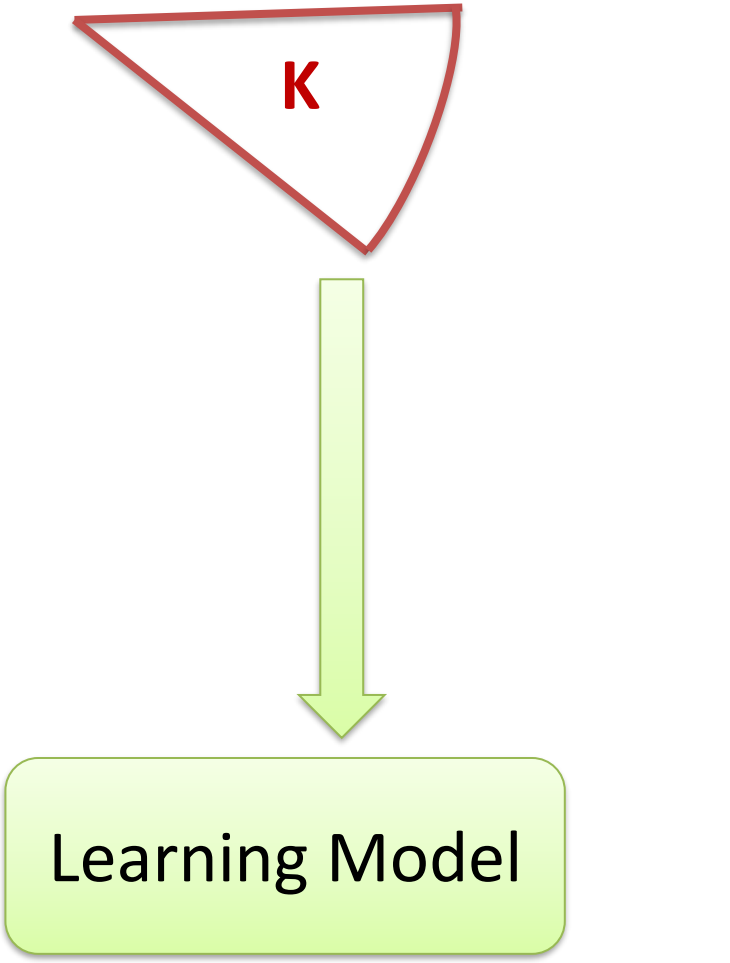
Learning Model



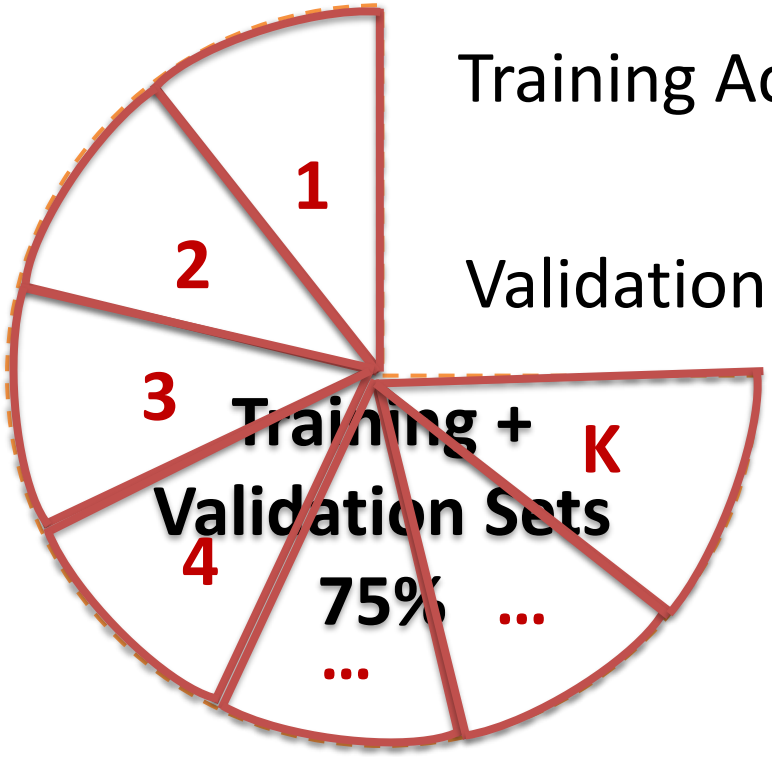
Learning Model

Training K Accuracy/Error


1
2
3
4
Training +
Validation Sets
75%
...
...

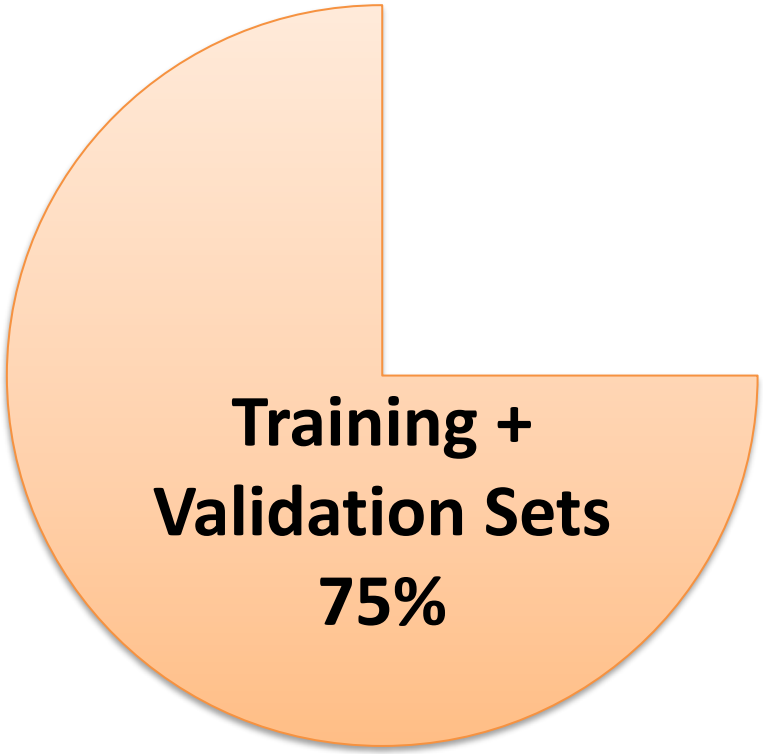

K
Learning Model
Validation K Accuracy/Error

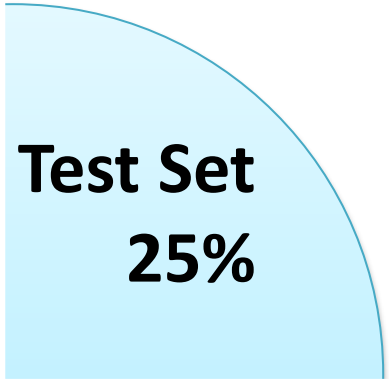
ทุก fold ที่ไป Test มาหาค่าเฉลี่ย

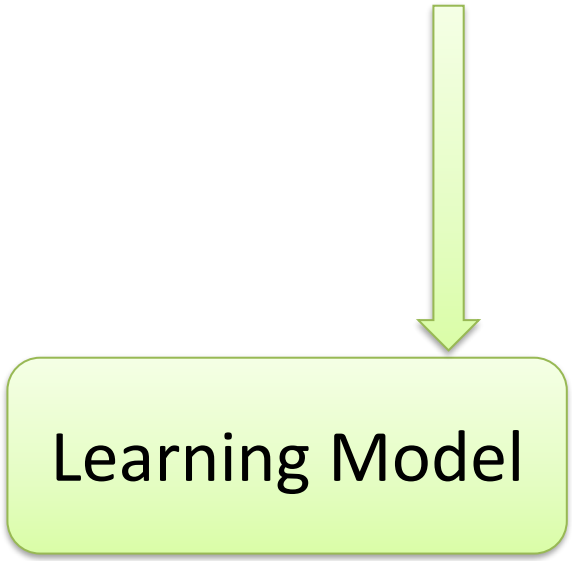


Training Accuracy = $(\sum_{i=1}^{K} Train_i_Acc)$/K

Validation Accuracy = $(\sum_{i=1}^{K} Valid_i_Acc)$/K







Test Accuracy/Error

Note: Percentage of test set can be varied, depends on how many total of instances we have.

Deep learning

(from wikipedia)

Deep learning (also known as **deep structured learning** or **hierarchical learning**)

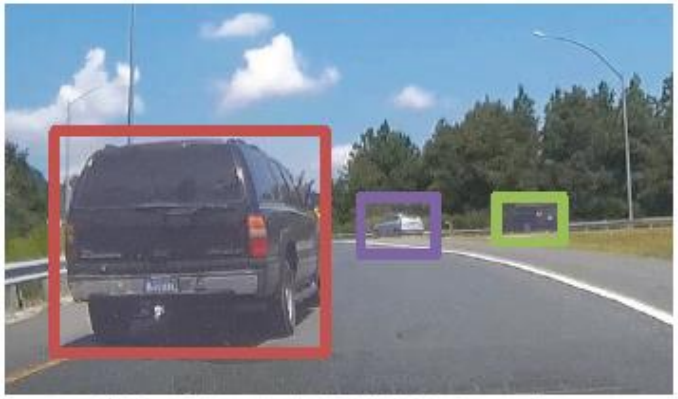
- Use artificial neural networks (ANNs) to learning tasks that contain more than one hidden layer.
- Deep learning is part of a broader family of machine learning methods based on learning data.
- Learning can be supervised, partially supervised or unsupervised.

Most popular applications of deep learning is computer vision.

Other applications are speech recognition, natural language processing, audio recognition, social network filtering, machine translation and bioinformatics.

This course will mainly focus on the convolutional neural network for the classification problem on image datasets.

There is a number of visual recognition problems that are related to image classification, such as object detection, image captioning.



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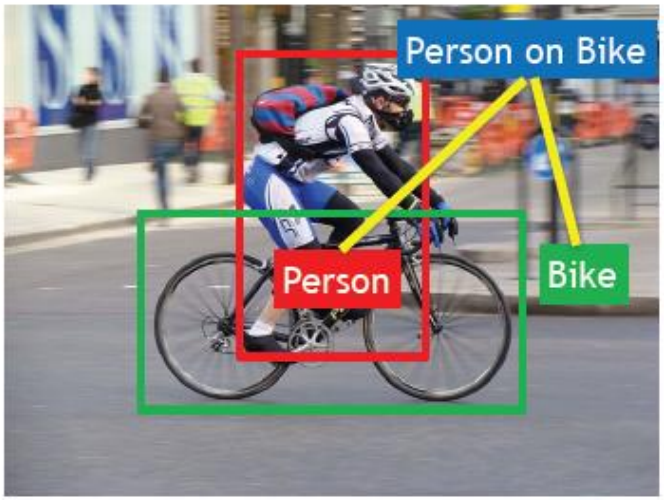
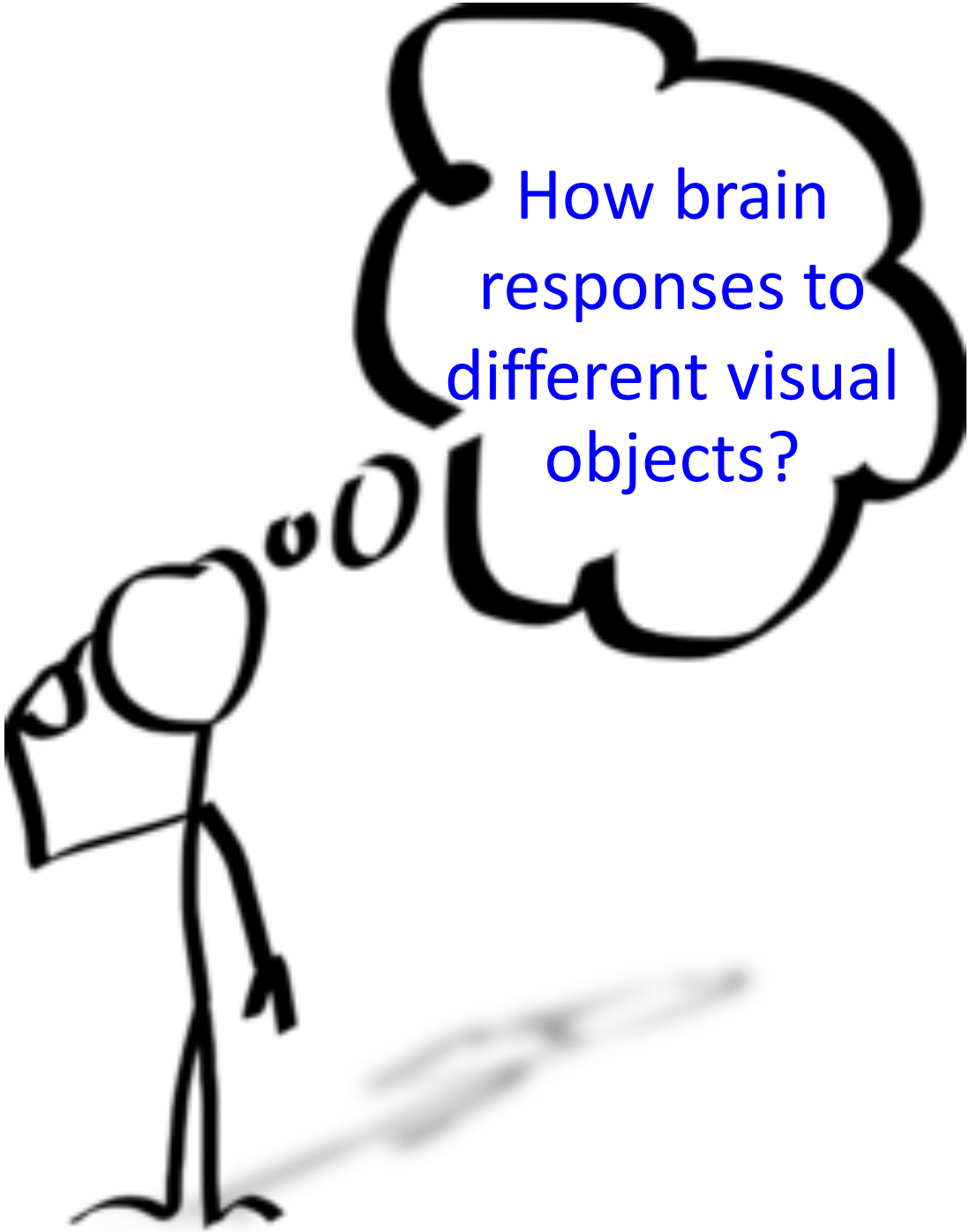
https://youtu.be/yQwfDxBMtXg

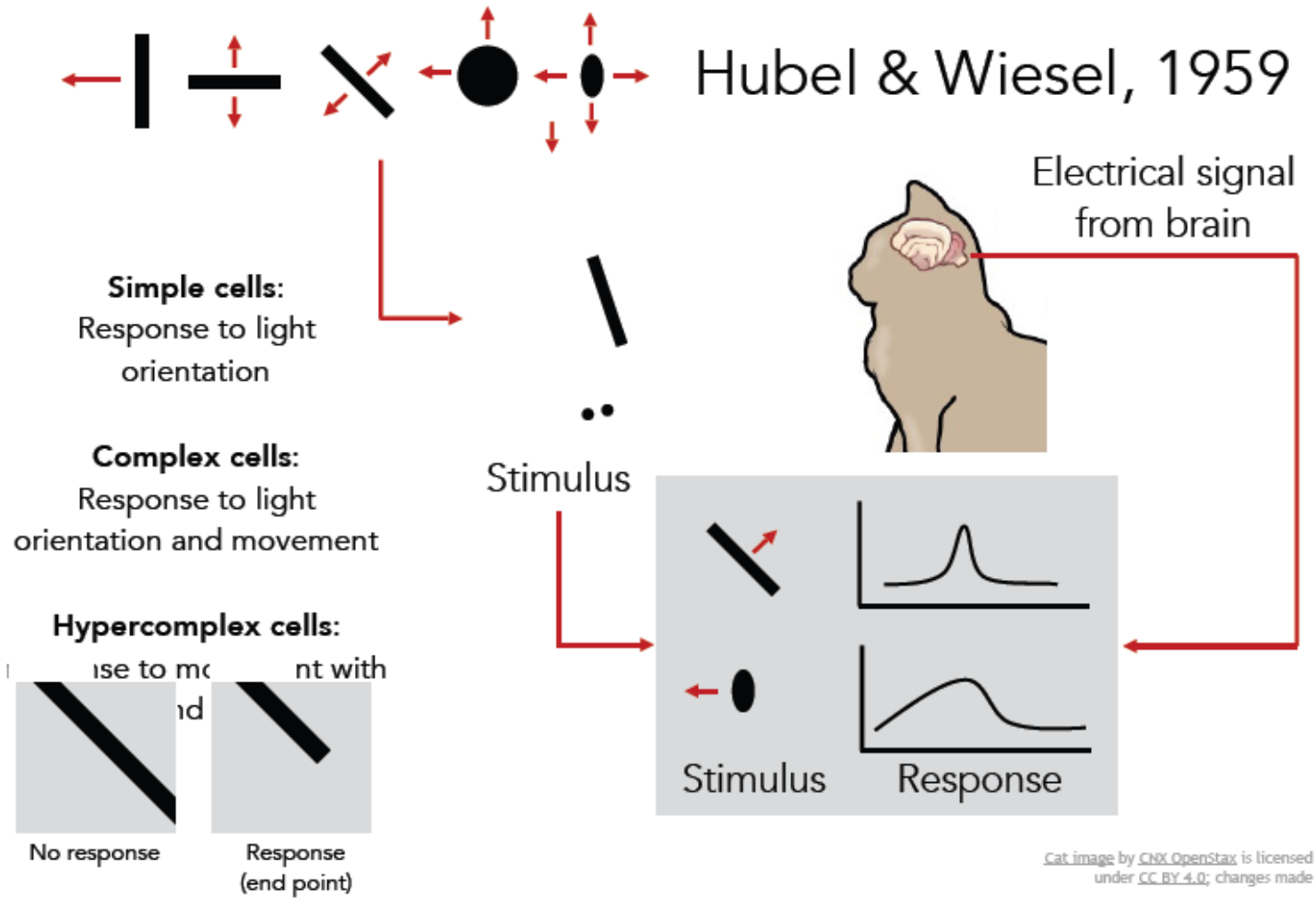
- Object detection
- Action classification
- Image captioning
- …



Images from cs231n @Stanford


How brain responses to different visual objects?

Visual Cortex Study



Cat image by CNX OpenStax is licensed under CC BY 4.0; changes made

https://youtu.be/IOHayh06LJ4

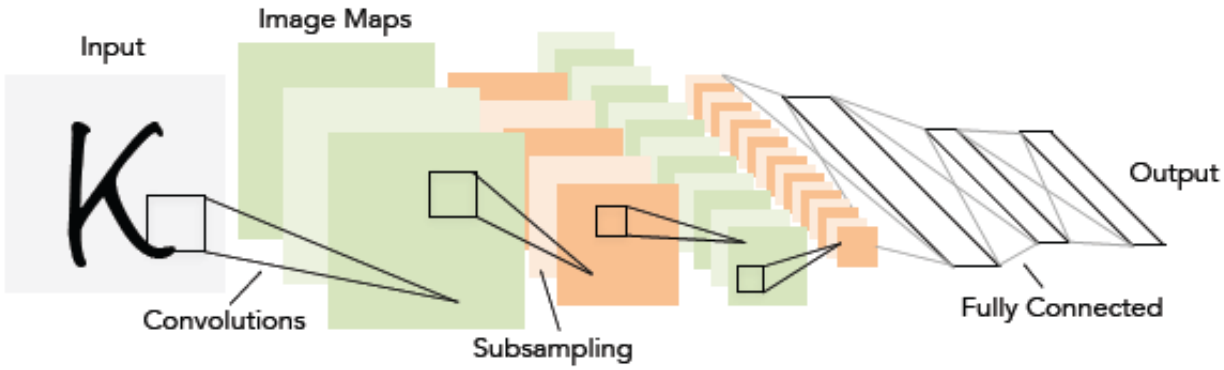
https://youtu.be/jw6nBWo21Zk

As you will see later in the class, convolutional neural network imitates visual cortext in the brain

Evolution of Convolutional Neural Networks (CNN)

1998

LeCun et al.



# of transistors

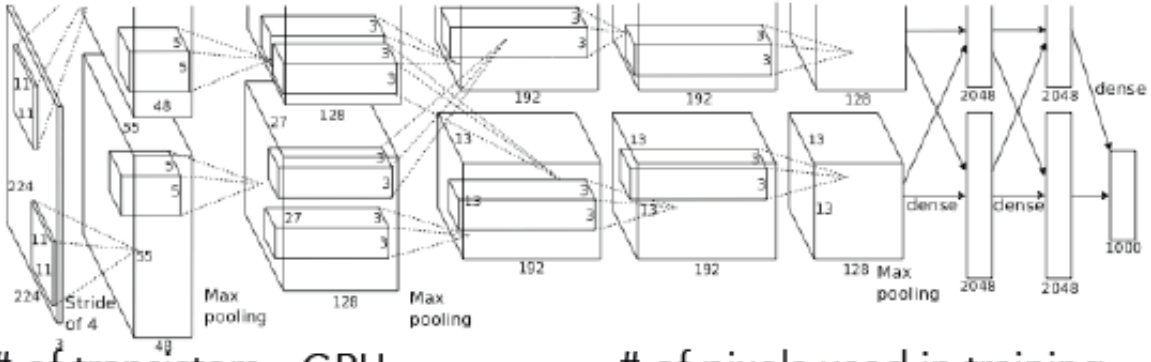
10^6

# of pixels used in training

10^7 NIST

2012

Krizhevsky et al.



# of transistors

10^9

GPUs

# of pixels used in training

10^{14} IMAGENET

Figure copyright Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, 2012. Reproduced with permission.